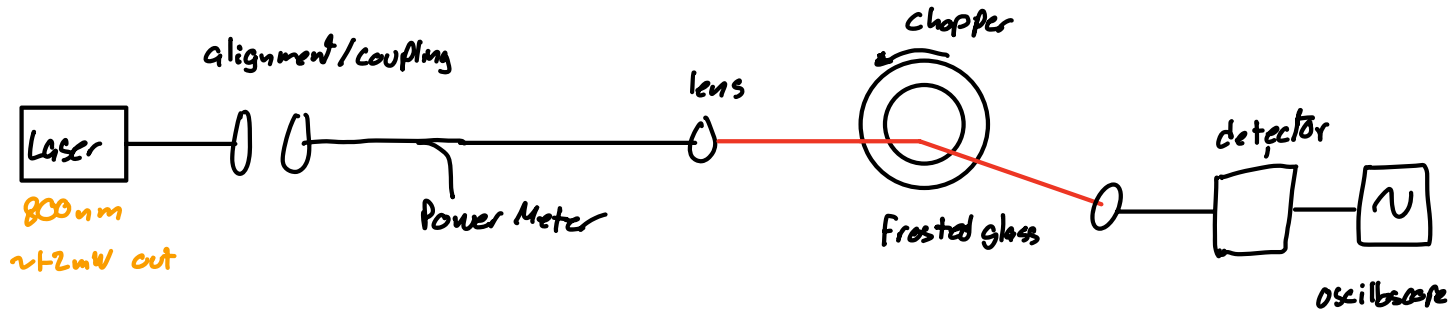
Photon Statistics Lab Notes:

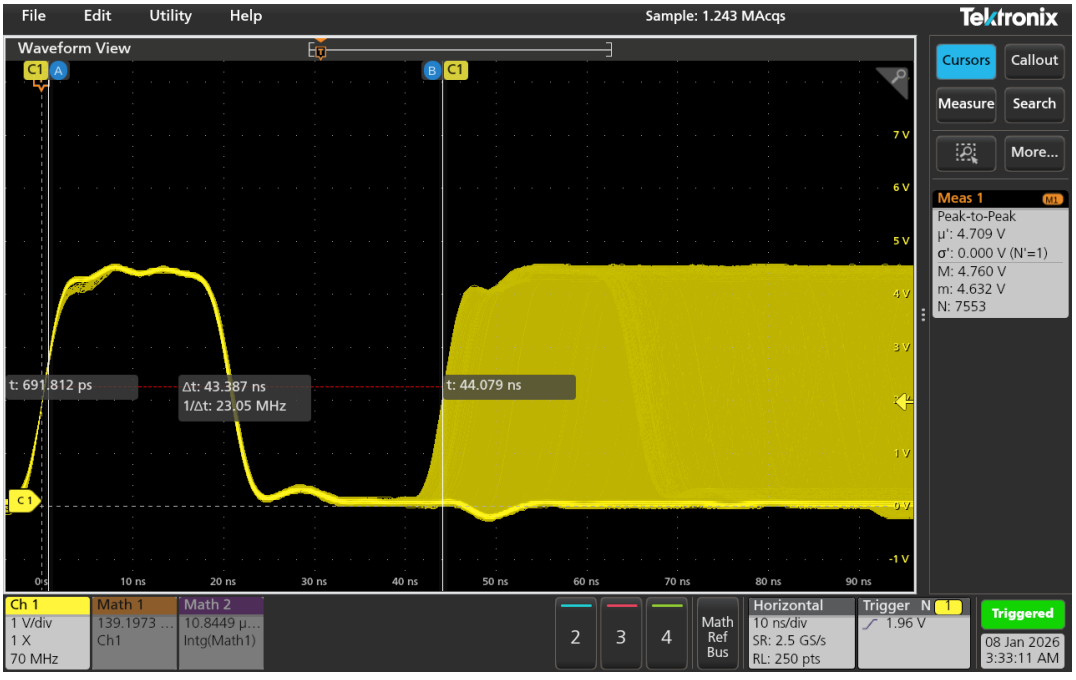
In the photon statistics station of this course, our goal is to use single photon detectors to explore the underlying statistics of detection events for various kinds of light sources. This will allow us to characterize (and differentiate) different light sources by analyzing their count statistics and multi-photon detection correlations. We will characterize our detector and photon count signal (signal rise time and detector dead time) and subsequently will investigate the count statistics for dark counts and coherent light source including the poissonian error model that underlies (and is often assumed) in their distributions. We will also investigate and characterize multi-photon event correlations for various light sources and characterize our detector's jitter which limits the temporal resolution of our detector.

Lab Day 1

In today's lab we performed experiments 1 and started experiment 2. In experiment 1 we characterize both the photon count event signal as well as the detector dead time. The count event signal follows the TTL standard meaning that it can be in a low voltage state (near 0V) or a high voltage state (~3V). Our experimental set-up is as follows:



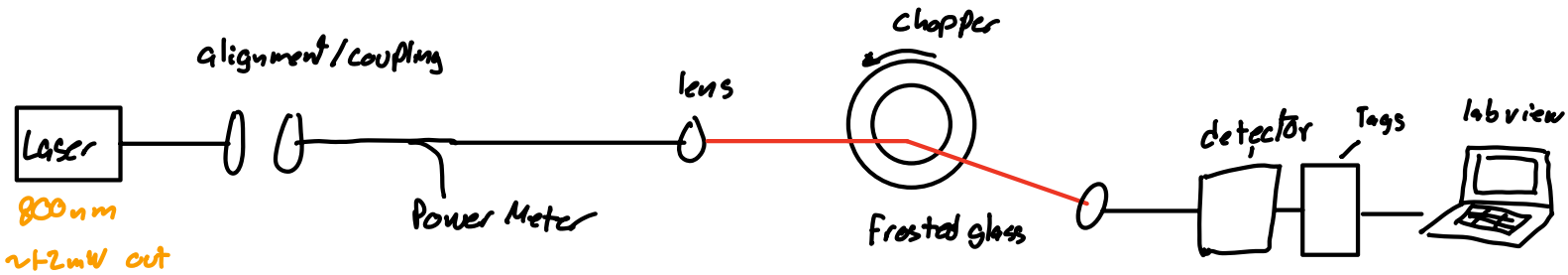
We first make sure our laser is properly coupled to our fibre. We do this by directly measuring the power out of the fibre using a power meter. We measured this to be 0.637+-0.0005 mW. We then let our light through to bounce of the chopper (not rotating in this experiment) to reduce the power of the light going into the detector to an appropriate level that won't damage it. Then we pass the signal to our oscilloscope to analyze our signal.



In the image above we characterize our detector deadtime and signal (which follows TTL standards as mentioned above). We measure a detector deadtime of 43.387+-0.065ns (our cursor precision was dt=0.092ns) and a rise time for our signal of 5.669+-0.065ns.

Experiment #2: Error Model

In this experiment we investigate the square-root-N error model (i.e. poissonian error model).



Here our experimental set-up stays mostly the same but now instead of sending our signal to the oscilloscope we send it to the time tagger and then to LabView. The relevant parameter for our LabView interface in this experiment is the integration time (the dt over which we count photons). To investigate the error/statistical model, we get 2 datasets.

First we use dark counts and acquire N data points for a given integration time. We repeat this for 3 different integration periods (1s,2s, 5s). We assume a poisson distribution meaning that the square-root of our mean should be equal to the standard deviation of each sample. The reason for this is we have a random process with independent events, occurring at some fixed rate over time, and where two events cannot happen simultaneously.

The 2nd experiment we basically repeat this experiment with the laser light on, this time varying the intensity of the light in the fibre by adjusting the coupling between it and the laser.

For the dark counts we have 3 datasets as follow:

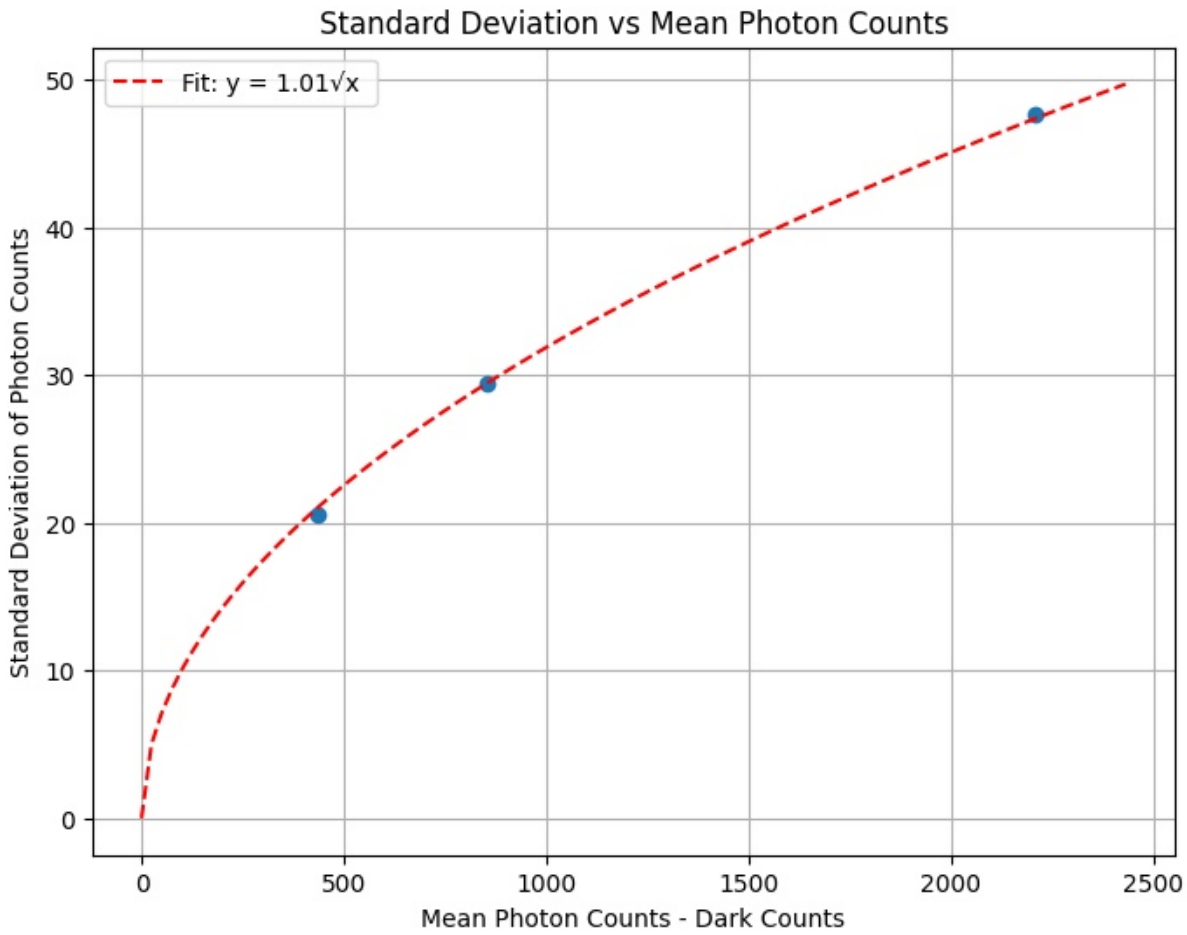
1s integration:
[420,450,400,457,416,465,398,418,415,434,401,415,429,469,454,409,453,432,456,450,426,441,439,444,424,444,458,408,451,457,440,476,442,441,448,434,431,464,427,479,482,422,451,425,433,423,442,467,418,436,404,441,444,457,464,439,412,431,490,430,422,427, 417,428,400,416,433,425,449,445,428,420]
N=72
Mean = 436.611
Std Dev = 20.559

2s integration:
[845,843,814,846,831,852,855,834,863,856,853,854,803,872,844,820,850,824,886,826,836,832,809,846,851,823,819,846,897,925,890,855,843,883,843,857,878,884,852,907,849,872,819,840,876,849,860,908,861,826,829,887,826,881,841,821,860,857,904,800,839,873,833,920,912,849,823,851,848,895,906,861, 845,832,871,889, 792, 874, 902, 897]
N=80
Mean = 855.3125
Std Dev = 29.384

5s integration:
[2181,2142,2224,2202,2158,2181,2199,2192,2209,2159,2141,2235,2181,2226,2325,2222,2256,2217,2186,2231,2218,2243,2281,2277,2185,2222,2231, 2278, 2284, 2145, 2118,2171,2187,2225,
2256,2192,2202,2120,2267,2157,2173,2188,2200,2299,2249,2162, 2277,2165,2174, 2165, 2181,2261,2258,2158,2169,
2223, 2246,2241, 2240, 2273, 2186, 2299, 2143]

N = 63
Mean = 2210.413
Std Dev = 47.627

Plotted:



For the laser light, we use an integration time of 2s:

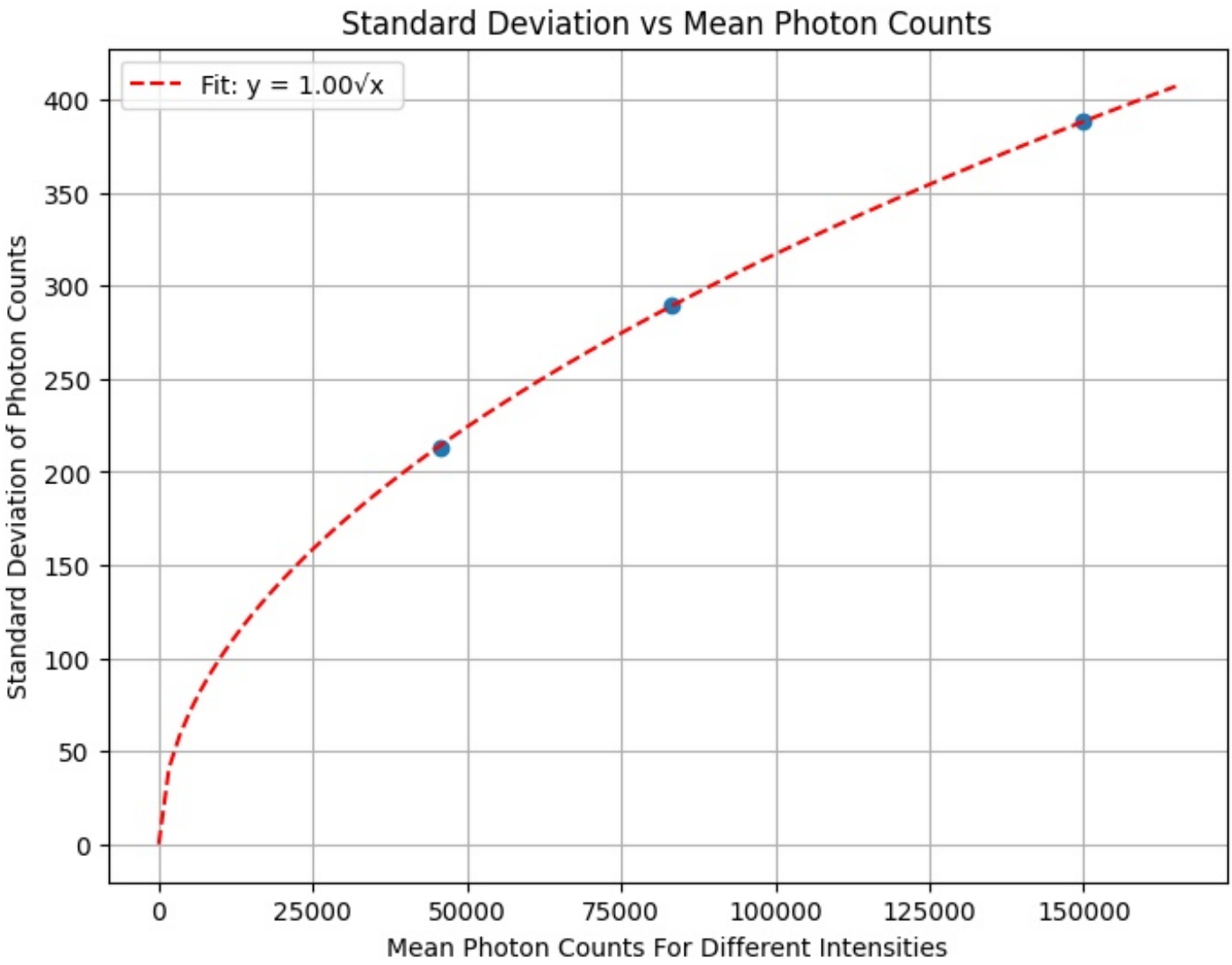
Lowest intensity
[45655,45777, 45987, 46066, 45621, 45549, 45833, 45599, 45739, 45525, 45681, 45673, 45565, 45460, 45620, 45077, 45288, 45708, 45313, 45274, 45477, 45632, 45457, 45272, 45607,45835, 45860, 45868, 45784, 45759, 45517, 45801, 46065, 45768, 45579, 45600, 45549, 45579, 45579, 45857, 46386, 45672, 45774, 46011, 45749, 45760, 46142, 45621, 45535, 45695, 45878, 45815, 45819, 45913, 45899, 45472, 45706, 45710, 45778, 45757, 45777, 45836, 45427, 45670, 45849, 45438, 45471, 45982, 45821, 45887, 45386, 45991, 45772, 45622, 45865, 45895, 45959, 45936, 45780, 45789, 45781, 45697]
N =82
Mean = 45709.49
Std Deviation = 213.53

Medium Intensity
[83299, 83423, 83416, 82528, 83470, 82943, 83149, 83615, 83343, 83424, 83299, 83290, 83200, 83519, 83482, 83564, 83366, 83404, 83367, 82927, 83507, 83457, 83393, 83421, 83445, 83267, 83616, 83174, 83072, 83578, 83411, 83555, 83077, 82822, 83115, 83295, 82344, 83533, 82515, 84078, 82935, 83241, 83185, 83291, 83294, 83310, 83198, 83243, 83209, 82909, 82881, 82921, 83130, 82700,83036,83013, 82929, 83438, 82929, 83318, 83345, 83085, 83426, 83585, 83523, 83125, 83319, 83040, 82940]
N = 69

Mean = 83234.797
Std Dev = 289.602

High Intensity
[149617, 149632, 150095, 150422, 150779, 149854, 149451, 150295, 149476, 149594, 149680, 150503, 150038, 150068, 149800, 149808, 150107, 150355, 150283, 150283, 150262,149625, 150716, 150157, 149560, 149930, 149575, 149922, 149221, 149911, 149361, 150009, 150191, 149643, 149877, 148742, 149939, 150023, 149727, 149830, 149732, 149991, 149636, 150286, 149846, 150287, 149768, 150340, 150400, 150164, 150337, 150602, 150400]
N = 53
Mean = 149965.094
Std Dev = 388.647

Plotted:



What we can see from these plots is that indeed our assumed poisson model is valid even under varying intensity and integration periods because the square-root of the mean is equal to the standard

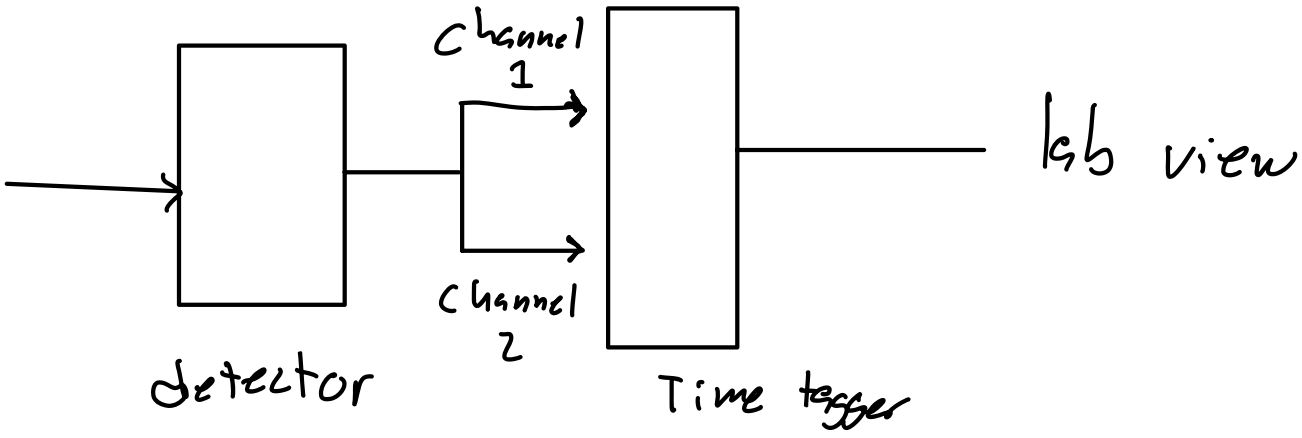
deviation for each of acquired datasets.

It's important to note that the data for the laser on experiment was taken with the chopper on (100+-0.5Hz) to reduce the variability of the environment since our light source reflects of the chopper therefore if the chopper rotates even very slightly (due to air currents or , so rotating this helps to mitigate this variability.

Experiment #3

In this experiment we explore measure the coincidence window, which is the window of time in which we consider 2 counts to be correlated. In other words any count where t+-dt/2 where dt is the coincidence window and t is the time or arrival of the photon, then we count these as correlated/ coincident. We measure this because we need this quantity to measure the 2nd order correlation function and it's value for tau=0 which we will use to characterize different sources of light.

Our set-up is practically the same as before for measuring coincidence windows with one difference, we split the output of the detector using a BNC Tee to split the single output from the detector to 2 input channels in the time tagger.



We change the settings in lab-view for each channel to being half-TTL (this tells lab-view to use half the voltage of the TTL standard to count, since we are splitting our signal in half, we're also splitting the voltage in half). We also make sure to count each channel, and to count coincidences between each channel. We vary the delays for each channel to measure our coincidence window (we are looking for when our coincidence counts drop to less ~1/2).

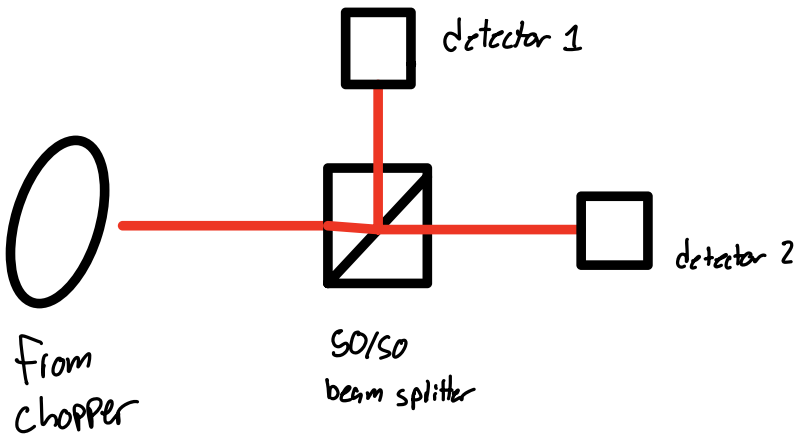
Max coincidence:
channel 1 delay = 2.96ns+-0.0775ns
channel 2 delay = 0ns+-0.0775ns

Half-max coincidence 1:
channel 1 delay = 6.08+-0.0775ns
channel 2 delay = 0+-0.0775ns

Half-max coincidence 2:
channel 1 delay = 0+-0.0775ns
channel 2 delay = 0+-0.0775ns

Therefore our coincidence window is 6.08+-0.1096ns

The 2nd part of our experiment is to measure the 2nd order correlation function at tau=0 g2(0) for laser light and pseudo-thermal light. We do this by assembling a Hanbury-Brown and Twist interferometer, so our set-up changes for the remaining experiments.



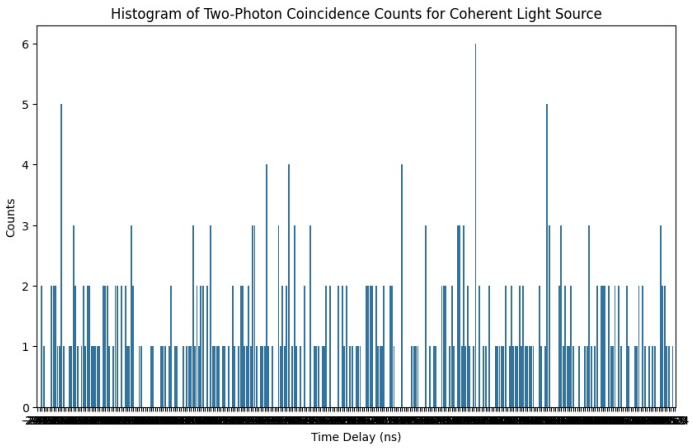
For the coherent and pseudo-thermal light sources, we use the following formula to calculate the g2(0).

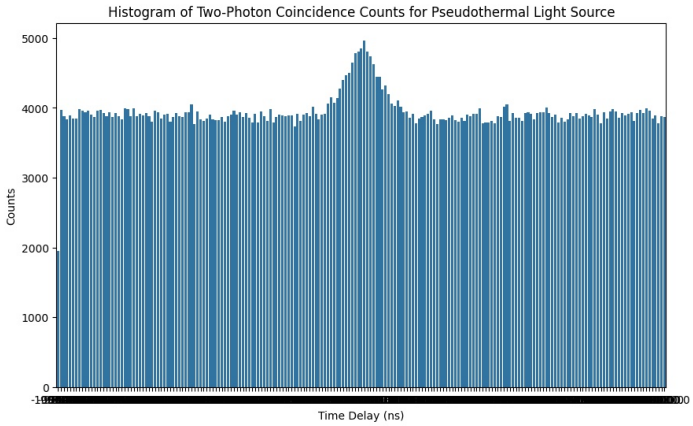
```
def g2(T, Nc, N1, N2, W):
    return (T*Nc)/(W*N1*N2)
```

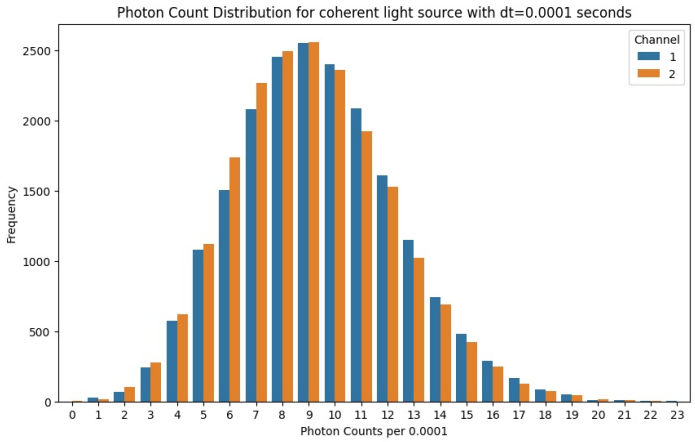
where T is the integration time, W is the coincidence window, Nc is our coincidence count, N1 is and N2 are the counts from each detectors respectively.

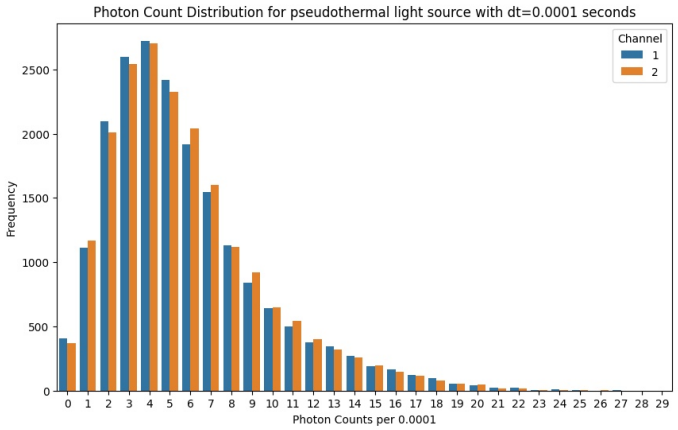
What we expect is a value of 1 for coherent light/purely poisson statistics. We expect a value >1 for pseudo-thermal light and bunched statistics. Finally for quantum sources of light we expect anti-bunching and a value <1.

In this experiment we also analyze the behaviour of g(tau) by looking at the histogram of our coincidence counts (varying tau does should not impact our N1, N2, W and T values so g(tau) is propotional to Nc(tau)). We also look at the frequency distribution of counts which should be poisson for coherent light and super-poisonian for pseudo-thermal light (bunched)









We see that the coincidence counts for coherent light is mostly constant no matter the value of tau, while the coincidence counts for pseudothermal light is roughly gaussian centered at 0, with a standard deviation determined by the rotation speed of the chopper. This is a bit technical to explain, but essentially the laser hitting the chopper has a gaussian profile. At a short delay tau later, the light is hitting the chopper in a slightly different place as the wheel has rotated. Our coincidence counts then are in essence measuring how much these two surfaces overlap (and since both are gaussian their overlap also behaves as a gaussian).

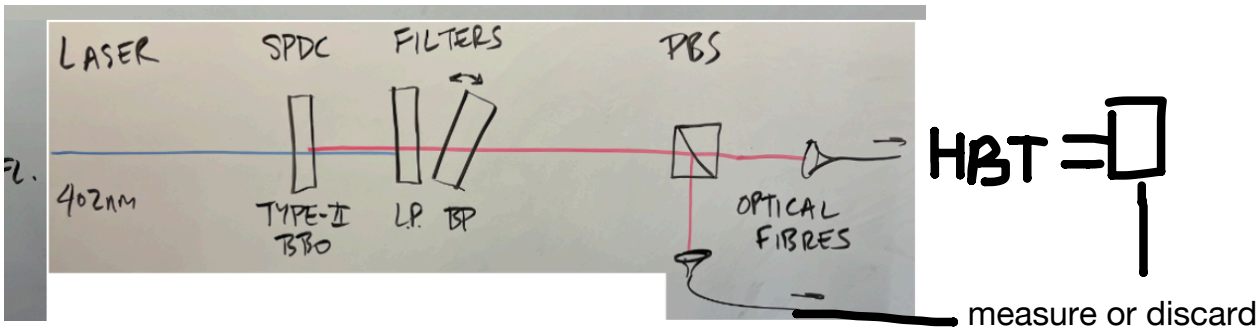
We choose an integration time of 300s to get good time averaging. We get the following g2(0) values:

Coherent light: g2(0) = 1.01+-0.01
with
N1 = 30492053
N2 = 31259926
Nc = 19555
This experiment should have been repeated for various intensities of light but ran out of time, additional measurements will be made for the post lab analysis.

Pseudo-thermal light: g2(0) = 1.53+-0.02
with
N1 = 16788276
N2 = 17369038
Nc = 9085
of note is that to generate our pseudo-thermal light source we rotate the chopper, we chose a value of 140+-0.5Hz as our rotation speed per the lab manual. We should have repeated this measurement for various rotation speeds, but we ran out of time. Additional measurements will be made for the post lab analysis

Experiment #4

This experiment is really similar to experiment 3 with the main difference in our set-up being our light source, as we now use a single photon source prepared via spontaneous parametric down conversion. SPDC works by passing high energy photons (in our case 402nm light) into a special type of crystal called a non-linear chi^(2) crystal. In our case this is a Barium-Borate (BBo) crystal. This crystal effectively can absorb the high energy blue photon and spontaneously emit 2 lower energy red photons (804nm). The reason we use Barium-Borate crystal is because of it's birefringent properties allowing us to do phase matching which allows us to tune the momentum-conservation of resulting photons to optimize the low energy photon pair production. Bellow is a schematic of the single photon source used



In the experiment we analyze g2 for triggered and untriggered SPDC. For untriggered, we essentially don't count the photons arriving to one of the detectors (effectively tracing out one of the qubits in our system), and for the triggered case we include the photons we previously discarded for the untigered case and use it as a trigger to count our 2 and 3 fold coincidences to measure g2(0).
we use the following function to measure the 2nd order correlation function valued at 0 for triggered light and the equation from experiment 2 for untriggered light
g2(0) = (Nt*N12t)/(N1t*N2t)

where Nt is the count of trigger photons, N12t is the 3 fold coincidences, N1t and N2t are the 2 fold coincidences between the trigger and our respective channels from the HBT interferometer. ideally we would have also taken this measurement for different levels of intensity but we ran out of time, I'll make sure to get these measurement for the post lab.

What we expect is a value <1 for triggered correlations, and for untriggered light >1

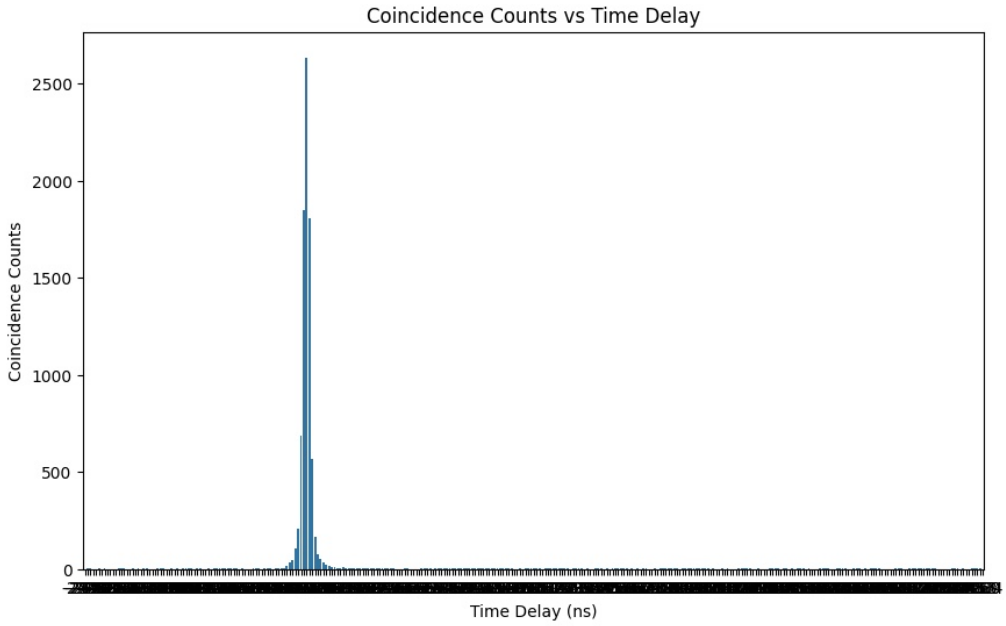
We get a g2(0) of 1.97+-0.03 for untriggered light we use the
Nc = 20999
N1 = 20440407
N2 = 25658522

and for triggered light we get g2(0) = 0.0178+-0.0004
Nt = 61734854
N1t = 2584187
N2t = 2846868
N12t = 2127

These results agree with what we expect for these sources of light, with our triggered experiment giving us a value for g2 consistent with anti-bunched statistics as we expect.

Experiment #5

Experiment 5 is quite straight forward, here we try to quantify the uncertainty in our time based measurements (also known as detector jitter) which limits the temporal resolution we can achieve. We measure the jitter of the system by measuring the coincidence between our trigger and one of the photons that passes through the HBT interferometer. This allows us to characterize the jitter by finding the FWHM. However it's important to note that the data we get is the jitter for 2 detectors, so to find the single detector jitter we assume both detectors are identical and have gaussian jitter and deconvolve the signal we observed from our raw dataset. The width of the resulting gaussian is then the jitter for a single detector.



Additional Notes

Important Files:

photon_counts.ipynb: has relevant code from the prelab and lab to measure our g2 functions, their uncertainty, plot our various plots, and our histogram generating code.

[**laserTAGS/pseudothermalTAGS].txt:** time tagger file for experiment 3 data used to plot our frequency distributions

g2laser.txt/g2pseudothermal.txt/counts.txt: histogram of coincidence counts for g2(tau) analysis.